

Slip 2

Q.1) Write a C program to find file properties such as inode number, number of hard link, File permissions, File size, File access and modification time and so on of a given file using stat() system call. [10 Marks]

```
#include <stdio.h>
#include <stdlib.h>
#include <sys/stat.h>
#include <unistd.h>
#include <time.h>

int main(int argc, char *argv[])
{
    struct stat fileStat;

    if (argc != 2) {
        printf("Usage: %s <filename>\n", argv[0]);
        exit(1);
    }

    // Call stat()
    if (stat(argv[1], &fileStat) < 0) {
        perror("stat");      // Print error message
        exit(1);
    }

    // Print file details
    printf("File Name: %s\n", argv[1]);
    printf("Inode Number: %ld\n", (long)fileStat.st_ino);
    printf("Number of Hard Links: %ld\n", (long)fileStat.st_nlink);
    printf("File Size: %ld bytes\n", (long)fileStat.st_size);

    // File Permissions
    printf("File Permissions: ");
    printf( (fileStat.st_mode & S_IRUSR) ? "r" : "-");
    printf( (fileStat.st_mode & S_IWUSR) ? "w" : "-");
    printf( (fileStat.st_mode & S_IXUSR) ? "x" : "-");
    printf( (fileStat.st_mode & S_IRGRP) ? "r" : "-");
    printf( (fileStat.st_mode & S_IWGRP) ? "w" : "-");
    printf( (fileStat.st_mode & S_IXGRP) ? "x" : "-");
    printf( (fileStat.st_mode & S_IROTH) ? "r" : "-");
    printf( (fileStat.st_mode & S_IWOTH) ? "w" : "-");
    printf( (fileStat.st_mode & S_IXOTH) ? "x" : "-");
    printf("\n");

    // Access and Modification times
```

```

    printf("Last Access Time: %s", ctime(&fileStat.st_atime));
    printf("Last Modification Time: %s", ctime(&fileStat.st_mtime));

    return 0;
}

```

Q.2) Write a C program that catches the ctrl-c (SIGINT) signal for the first time and display the appropriate message and exits on pressing ctrl-c again.

```

#include <stdio.h>
#include <stdlib.h>
#include <signal.h>
#include <unistd.h>

int sigint_count = 0;

void handle_sigint(int sig)
{
    sigint_count++;
    if (sigint_count == 1)
    {
        printf("\nCaught SIGINT (Ctrl-C) for the first time! Press Ctrl-C
again to exit.\n");
    }
    else
    {
        printf("\nExiting program after second SIGINT (Ctrl-C).\n");
        exit(0);
    }
}

int main()
{
    // Set the signal handler for SIGINT
    signal(SIGINT, handle_sigint);

    printf("Press Ctrl-C to test the signal handling.\n");

    // Keep the program running to catch signals
    while (1)
    {
        pause(); // Wait for a signal
    }
}

```

```
    return 0;  
}
```